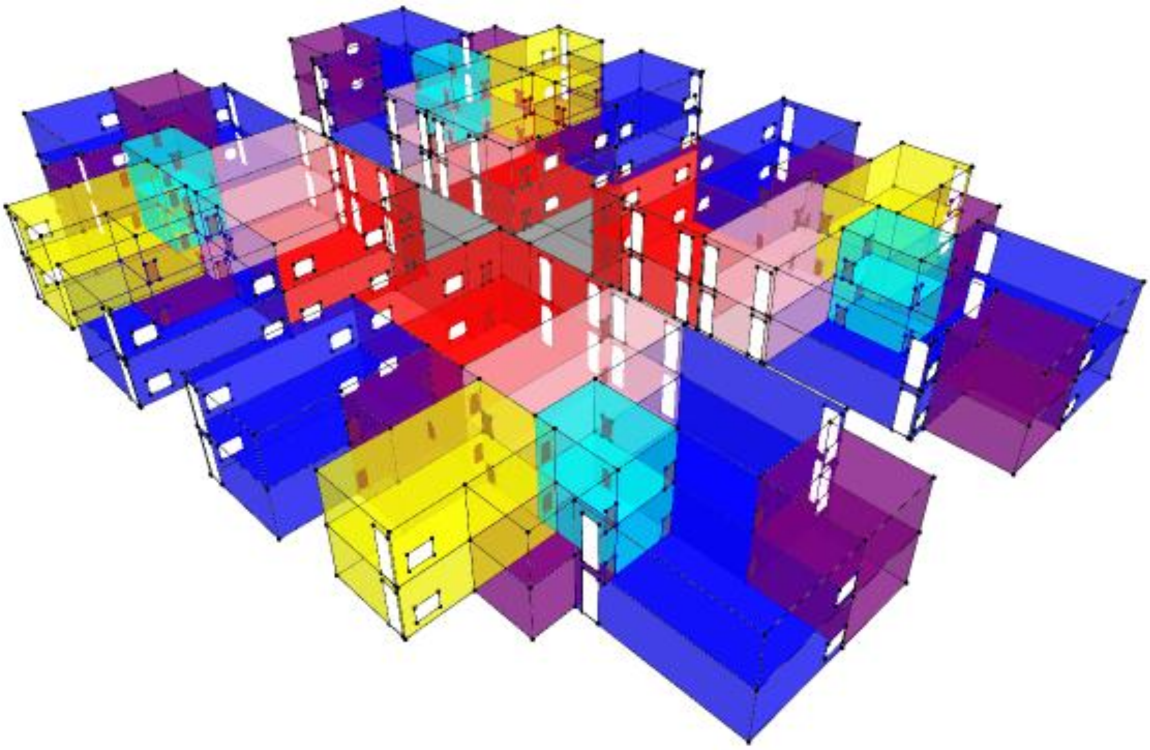
Apartment Complex Network Graph

This project uses topological modelling to translate the spatial organization of an apartment complex into a graph, revealing patterns of connectivity, accessibility, and spatial hierarchy within the residential layout. The visualization abstracts physical spaces into nodes and their relationships into edges, allowing the building to be understood as a network of interconnected living environments.



<u>how architectural elements are mapped to graph components:</u>

Nodes (vertices) represent distinct spatial units within the apartment complex. These include individual apartments, shared corridors, staircases, lobbies, and service areas. Each node is encoded with attributes such as function or usage type and its size reflects the relative area or importance of the space.

Edges represent relationships between these spaces. The primary connections correspond to direct physical adjacency—such as doors linking apartments to corridors or corridors to vertical circulation cores. Secondary connections may represent visual links or shared access zones, such as balconies or openings that connect interior spaces to the exterior environment.

the logic behind node and edge definitions:

The graph structure illustrates how residents navigate through the complex, highlighting both direct and indirect paths between spaces. Central nodes typically correspond to shared circulation areas, acting as hubs that connect multiple apartments. Peripheral nodes represent more private residential units, emphasizing their relative isolation within the network.

By mapping adjacency and access as edges, the model exposes the hierarchy of movement—from public entry points to private living spaces—and identifies critical connectors that influence flow and accessibility. The resulting network makes the spatial logic of the apartment complex legible, offering insights into circulation efficiency, clustering of units, and the balance between shared and private domains.